

Work Breakdown Item	Front end Hours		Back end Hours		ML/AI Hours		UI/UX Hours		Stage MVP	Development comments	Features comments
Min	Max	Min	Max	Min	Max	Min	Max				
Base project setup											
UI Kit							24	40	✓		
Project setup	8	16	8	16	4	8			✓		
Architecture	16	20	16	20	4	8			✓		
Base components and UI theme	40	40							✓		
DB structure setup			24	32	4	8			✓		
Localization	44	60	26	40	12	16	12	16	✓	English and German, fixing prompts and FE logic, setting error codes on both languages	
Role model	24	32	24	32					✓		
Project features											
TECHNICAL PART											
Module 1: EHR Data ingestion, generation & NLP processing											
Automated incremental FHIR sync			24	32	8	12			✓	External API <-> BE -> ML -> BE -> FE -> BE -> ML	Data from the following resources should be captured: Patient, Encounter, Condition, Observation, DiagnosticReport , MedicationRequest/MedicationStatement, Procedure , AllergyIntolerance , FamilyMemberHistory, DocumentReference .
Normalization and provenance retention			12	18	32	40			✓	For example, https://loinc.org/xb/api/search-api/ can be used for cases where we need to extract code from free text(for guideline processes) and, since we don't have other data, we suggest to postpone the mapping with some internal clinical codes	The system must normalize clinical codes to SNOMED CT/LOINC/CD-10 and retain original codes and Provenance metadata (source, ID, timestamp, model version).
Entity extraction (NLP)			28	32	60	64			□	The integration with https://huggingface.co/emilyalsentzer/Bio_ClinicalBERT , and with https://huggingface.co/medicalai/ClinicalBERT , can be arranged, but we would need to discuss the data where it should be applied, since most commonly it's designed for the patient data, but maybe it can find a use for the guidelines as well	The system must extract clinically relevant entities (symptoms, medications, negation, temporality) from free text and assign an NLP confidence score.
Non-text data handling			4	8	2	4			✓		Process only free-text present in FHIR fields and text-based attachments; no OCR and no parsing of PDFs/images. The system must skip non-text documents (PDFs, images) and flag them as unsupported_nontext in the UI and telemetry.
Structured conflict reconciliation					12	16			□	For this cases we would need to create a mocked data pipeline, since in Synthea there is no conflicts of extraction, but with AI module for fhir generation it can be	The system must de-duplicate and surface conflicts between NLP-extracted data and existing structured facts from the EHR.
NLP fact visibility control	12	16	4	8	12	16			□	Applicable in this stage only with real data/AI generated patients data. But logical - all of the textual data will go insed of the model , if it was mentioned in the mandatory fields, probably will not make much of a difference	The clinician must be able to include/exclude NLP-extracted facts from diagnostic reasoning using UI toggles, with these facts being clearly badged ("NLP-extracted").
AI patient data generation	32	40	32	40	96	120			✓	We propose to create fhir-agent-generator with LLM connection and fhir schema creation, so by the disease definition we would be able to sample patients. As well attempt on creating Synthea json might be performed	This is not an EHR, but a temporary solution if there is no integration with real EHRs. Set of input fields to create a Patient data then to transfer to FHIR format.
Unified integration layer creation			24	64					✓		
Creating a common data collector			24	32	32	40			✓		Taking into account that you will need common statistics based on data from different clinics.
Module 2: Diagnostic engine & ML											
Pubmed Intergration - inital parsing and connection			40	48	80	96			✓		Download the inital most relevant data for the 250 diseases, parse and analyze, create a flow engine
Pubmed Intergration - automated updates			20	32	40	64			✓		Intergrate the previous engine on autonomus level, created cron jobs for the updates, validate

Done

Backend

Problems

To do ML

to do

(now impossible)

(now impossible)

(now impossible)

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Patient data analysis and diagnosis assumption with the confidence score			4	16	104	128			☑	On the ML side - refine the engine for 250 diseases, switch to graph based knowledge system	The system must produce a ranked list of candidate rare conditions for the patient with confidence scores. Hematology / Oncology (~250 conditions) + IMCD (and related conditions)	
Evidence panel for each suspected condition	6	8	8	12	32	40			☑		The system must display evidence (supporting, inconsistent, unrelated) for each candidate diagnosis with links to the source FHIR facts.	
Differential diagnosis providing (Narrative)			4	12	16	24			☑		The system must generate a holistic narrative conclusion on top alternatives, listing key discriminators and factors of uncertainty.	
Structured patient summaries creation			4	12	6	8			☑	Include new data	Covering patient history, investigations, family history, and problems/medications	
Rate limit handling for manual creation/data updates	4	8	4	12	4	12			☑		Highlight server loadness and ask user to wait more	
Module 3: Analytics and validation module (Background data collection, no clinician UI required)												
3.1 Match capture engine												
Automatic match trigger			20	32	4	8			☑		The system must automatically activate the capture process when a confirmed diagnosis (FHIR Condition, verificationStatus=confirmed) appears in the EHR.	to do
Prediction matching rules			20	32	4	8			☑		The system must match the confirmed diagnosis against a prior RareSight prediction based on configurable rules: Top-3 prediction within 180 days of the prediction with a minimum confidence ≥50%.	to do
Time-to-diagnosis calculation			4	8	4	8			☑		The system must calculate and save the time-to-confirmed diagnosis metric (time from RareSight prediction to official EHR confirmation).	to do (connected with automatic match trigger)
De-identification pipeline			8	16	16	24			☑		The system must automatically trigger the de-identification pipeline for all captured records.	
3.2 Data repository structure and												
Anonymized insights repository			12	24	16	24			☑		Create a dedicated repository for analytics using a Parquet/columnar schema architecture.	More back
Key metadata recording			4	8	4	8			☑		The system must save for each record: disease code, prediction rank, confidence, prediction timestamp, and time-to-confirmed diagnosis.	
Clinical fact recording			4	8	4	8			☑		The system must record extracted signs/symptoms and key findings used by the model (without storing raw free text).	
Geography and demography recording			4	8					☑		The system must record the clinic/organization ID and region (no patient geolocation) and the patient's age band/sex.	to do (add to FHIR)
3.3 Compliance and confidentiality (anonymization and governance)												
k-Anonymity and cell suppression			8	16	8	16			☐		The system must apply k-anonymity rules (k≥5) and the small-cell suppression mechanism to all aggregated data.	to do (tests)
Purpose limitation			8	16	8	16			☑		The system must limit data use to aggregated insights only (no patient-level sharing) and ensure data minimization.	
Export control			8	16					☑		Develop export control mechanisms that will block data release that fails k-anonymity checks.	Backend
USER PART (note: only additional features for the existing interface were mentioned here)												
Module 1: Clinician dashboard & management												
1.1 Patients management												
Receive data about Patient with profile creation	12	16	8	12					☑		Described in details in EHR Data Ingestion & NLP Processing section.	
Patient data creation in case no integration with real data provided									☑	Assumption: Need to implement endpoints for managing all supported FHIR resources with their properties		
Highlight of missing/contradictory data	8	12	8	12					☑		Omitted critical info → "data completeness" banner.	empty EHR (to do)
Mark Patient as Favourite	8	12	4	8					☑	Already implemented in current version, but could change with database models change.	Logic of marking can be changed	
Structured patient summaries viewing	16	24	4	8					☑		Auto-summary of history, investigations, family history, meds/problems; action list with links to evidence.	
Diagnosis mismatch flags providing	8	12	8	12					☑		Patients where the EHR diagnosis is not equal that RareSight's prediction with a confidence of ≥70% should be highlighted/flagged.	

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Filtering and sorting of patients	32	40	40	64					✓	The current implementation of sorting, filtering and pagination must be carried out on backend	Sorting (multi-criteria, stable): by confidence score (desc/asc), disease/likely condition, flagged status, favorites, last updated / new data since last review, patient age. Filtering (combinable): - Clinical: therapy area (Hematology/Oncology), confidence ranges (e.g., ≥70%, 50–69%), flagged mismatch (EHR dx ≠ RareSight top dx & confidence ≥70%), new data present, data completeness (complete/missing key facts). - User-centric: favorites (starred), assigned to me (if assignment exists), review status (unreviewed/reviewed). - Demographics: age (bands & min/max), sex (if available from EHR).
Quick search	8	12	8	16					✓	Assumption: Simple SQL query search	Quick search by MRN and disease keyword.
Data state indicator	12	16	8	12					✓		Patient's last data update timestamp and a data staleness indicator should be displayed.
Saved view management	16	24	10	24					✓	Storing User's filters and sorting configuration presets	User must be able to save and load their custom filter/sort presets (Saved views).
1.2 Diseases management											
Guideline integration provided to have a disease description	6	8							✓	estimated ai/be in pubmed integration	
Sorting by name	2	4	1	2					✓		
1.3 In-app user guide											
User guide info updating	8	12							✓		
Module 2: Manual case											
Manual data entry form	32	40	24	32			8	12	✓	AI Side in AI patient data generation	Manual entering of clinical data, which is automatically converted to FHIR R4 for model analysis. No linkage to real patients/EHR charts: manual cases live in a separate Manual Case workspace; they do not appear in the clinical patient dashboard, do not update EHR data, and do not participate in mismatch flags or match-capture analytics.
PHI warning and isolation	8	16	8	16					✓		The system must display a persistent PII warning banner and strictly isolate manual cases from the clinical dashboard and analytics.
Template management	24	32	24	32					✓		Template management: ability to Save, Load, Rename, and Duplicate manual cases as templates (e.g., "Patient A"); templates are versioned and scoped to the user/tenant; optional "Favorite" flag.
Mandatory NLP suggestion confirmation	8	16	8	16					✓		For manually entered text, explicit clinician confirmation is mandatory before including extracted NLP facts in the analysis bundle.
PROCESSES PART											
Module 1: Compliance, Security & Analytics											
GDPR rules			8	16					✓		DPIA complete; data minimization; configurable retention/erasure; lawful basis & records of processing.
RBAC and audit			40	60					✓		Ensure clinician-only access (RBAC) and create immutable audit logs of all actions, model versions, and configuration changes. Add RBAC on DB level.
EU AI Act logging			24	32					✓		Must maintain logs of inputs, salient features, model version, monitoring plan and outputs in accordance with EU AI Act requirements.

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MDR documents preparation			16	32					☒		- Core Technical Documentation (MDR) - Intended Use & Claims (conservative, evidence-aligned) - System Architecture & Data Flows - GSPR Matrix (Annex I) - Risk Management File (ISO 14971): Hazard analysis, risk controls, residual risk, benefit-risk - Software Lifecycle (IEC 62304): SDP, SRS, architecture, unit/integration/system test plans & reports, anomaly log - Usability Engineering File (IEC 62366-1): Use spec, HRA, formative & summative test reports - Cybersecurity Plan: Threat model, SBOM, patching/update policy, secure config guide, tamper-evident logging - Model Governance Dossier: Data lineage, training/eval protocols, performance metrics, monitoring & drift plan, model card, versioning/rollback SOP - Clinical Evaluation Plan (Annex XIV): Strategy using retrospective analysis now; prospective plan outline - Post-Market Surveillance Plan: Metrics/events to collect during pilot; vigilance procedure - Labeling & IFU: On-screen labeling, arnings/limitations, training content; alignment to risks/usability - Traceability Matrix: Requirements ↔ Risks ↔ Tests ↔ Evidence
Release notes preparation									☒		
Pilot analytics									☐		Aggregated metrics on active users and statistics on suspected/confirmed iMCD cases (by confidence bands). Initial focus on iMCD Metrics to collect (minimum): - Active tool users - Location of tool users - Suspected iMCD patients by confidence - Location of suspected iMCD patients - Confirmed iMCD diagnoses (number & location) - Symptoms / clinical presentation statistics (iMCD) Creation of the Analytics interface: Developers must create internal dashboards to visualize metrics (active users, iMCD cases, etc.). Flexible Export: Provide the ability to export this data in standard formats (CSV/JSON). Filtering: Implement mandatory filters for data analysis based on time window, site/region, and confidence band. Configuration-Level Access Control: The IT team must implement a switch (feature flag/configuration) for each pilot site that allows the system to enable or disable the collection and display of metrics specific to that institution. This provides full control over adherence to local agreements. Delayed Data Collection Activation: The diagnosis match-capture mechanism must be initially disabled. It should only be activated approximately two weeks after the system goes live at the site. This requires the implementation of a timer or a manual, timestamped switch to ensure the gradual and controlled start of sensitive data collection.
Accessibility WCAG 2.1 AA standart	64	72					30	40	☐	Initially we didn't have a requirement to support web accessibility, we need to make changes in UI	
CDS Hooks Support <i>(Optional)</i>									☐		Specifications for CDS Hooks (patient-view, order-select) to surface suggestions directly in the EHR interface.
									☐		
Unit Tests	40	64	30	60	30	60			☒		
Adaptive Markup Implementation (web)	40	48					30	40	☐		
Error States and loaders	16	20	20	30					☒		
Project Technical Documentation	8	16	8	16	8	16			☒		

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Communication: Scrum (planning, daily, demo, retro), requirements clarifications	20	40	20	40	20	40	10	14	☒		The implementation effort presented also includes communication, as communication is inevitable, as individuals make up the team. Communication is the key to project success and the team members need to be involved in certain discussions, both planned and unplanned, among which are: - Daily Meetings - Sprint Planning - Sprint Review - Retrospective - Interteam communications (discussions on the matter of technical implementation, architectural choices, PR reviews, merge requests, as well as any meetings aimed at improving the processes, technical & project documentation, acquiring the understanding of the business part of the project).
Total Development	582	796	763	1,254	686	960	114	162			
Total MVP	466	660	723	1,198	594	848	54	82			